

CROWN ETHER ENHANCEMENT OF IONIC CONDUCTIVITY IN A POLYMER-SALT SYSTEM

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Crown ethers are a class of organic compounds that form complexes with inorganic cations. When crown ethers are added to poly(vinylene carbonate) containing dissolved lithium salt, ionic charge transport in the solid electrolyte is assisted. The ionic conductivity of the polymer containing 12-crown-4 is three orders of magnitude greater than the conductivity in the polymer without crown ether. The conductivity of this system at room temperature is approximately $2.5 \times 10^{-4} \text{ S cm}^{-1}$, higher than any polymer-lithium salt system yet reported. The effects of various crown compounds as well as their concentration effects are examined.

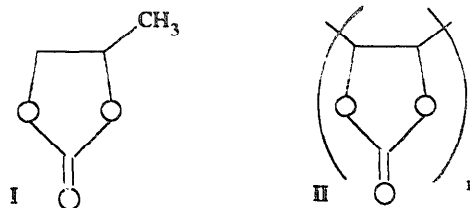
1. Introduction

The advent of microelectronics has stimulated research in the area of new lightweight power storage sources. Among the major desirable objectives for such high energy density battery systems are solid electrolytes that might be formed in very thin sheets thereby permitting flexible, large surface area electrodes to be used. Toward this end a number of polymeric systems have been examined worldwide.

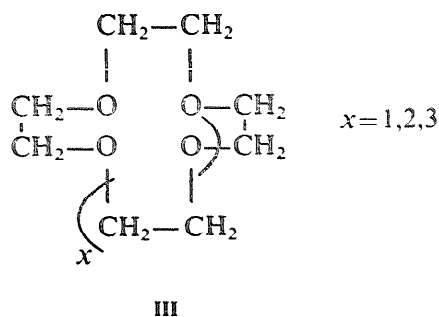
Ionic conductivity in alkali cation – poly(ethylene oxide) (PEO) complexes was first demonstrated in 1973 [1] and since that time numerous studies based on PEO have been reported. Unfortunately, the room temperature ionic conductivity of alkali cation – PEO complexes is not high enough to be of great interest for batteries ($< 10^{-5} \text{ S cm}^{-1}$). Efforts to improve ionic conductivity in polymer-salt systems generally fall into three categories. These are that (1) the PEO system is modified [2], (2) different types of polymers are synthesized [3] and (3) additives to PEO or similar systems are tried [4].

In this paper we report on our approach to enhancing conductivity using a polymer structurally similar to propylene carbonate (I) a common solvent used in non-aqueous electrolytes. The polymer,

poly(vinylene carbonate) [5] (II), was used in conjunction with various crown ethers [6] to study the electrical properties of their lithium salt complexes. When polymerized with a free radical initiator such as benzoyl peroxide, a hard clear plastic results, unlike PEO which is soft, translucent and rubbery. Crown ethers (III) are macrocyclic compounds in which ether oxygens inside of the ring



structure are perfectly arranged to permit metal ions to enter and interact (complex) with the oxygen lone pairs. Various ethers are known to form relatively strong complexes with specific alkali metal ions.



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2. Experimental

Vinylene carbonate (Aldrich Chemical Company, Milwaukee, Wisconsin) was distilled, in vacuo, from NaBH_4 and stored under nitrogen in a refrigerator until used for polymerization reactions. Samples for polymerization were prepared in a dry nitrogen glove-box and contained the requisite amounts of vinylene carbonate, LiCF_3SO_3 , crown ether and benzoyl peroxide initiator (see table 1). The samples, in ground glass-jointed test tubes with constrictions, were closed with a stopcock, removed from the dry box and subjected to three freeze-thaw vacuum degassing cycles. After the last pump down the tube was sealed by fusing the constriction and the resulting ampoule thermostatted at $73 \pm 2^\circ\text{C}$ until the mixture had solidified (16–40 h).

The sample tubes were returned to the dry box, opened and a portion of the clear, pale yellow plastic removed for electrical evaluation. Complex impedance measurements (from 40–125°C) were performed as previously described [7] and used to determine conductivity.

3. Results and discussion

The conductivity data versus temperature for the various crown ether samples and controls are shown in fig. 1. When only LiCF_3SO_3 (at the same concentration) is present in the poly(vinylene carbonate) (PVIC) the conductivity at $\sim 40^\circ\text{C}$ is $10^{-7} \text{ S cm}^{-1}$ and rises slightly until about 100°C when it then begins to decrease.

The family of curves that represent the PVIC-lithium salt complexes with added crown ethers exhibits a significant increase in conductivity over that of the sample without crown ether. For example, at $\sim 40^\circ\text{C}$ the observed conductivity for the 12-crown-4 sample is $2.5 \times 10^{-4} \text{ S cm}^{-1}$, or more than 3 orders of magnitude higher than the control. To our knowledge, this is the highest ionic conductivity yet reported in polymeric solids at or near room temperature.

The conductivity trend among the crown ether samples is 12-crown-4 > 15-crown-5 > 18-crown-6, with the cavity size increasing in the same order. The 12-crown-4 is lithium specific, that is, it does not complex metal ions that are larger than lithium. In this case only 1:1 complexes, which are the most common, are considered. At first glance it might seem that lithium specific complexes should decrease con-

Table 1
Sample composition for polymerization ^{a)}.

Additive (M.W.) (cavity, Å) ^{b)}	Moles	Moles vinylene carbonate	Moles LiCF_3SO_3	Monomer Li
12-crown-4 (176.2)	0.002	0.0398	0.002	20
(1.2–1.5)	0.0004	0.0398	0.002	20
15-crown-5 (220.3)	0.004	0.0398	0.002	20
(1.7–2.5)	0.002	0.0398	0.002	20
18-crown-6 (264.3)	0.002	0.0398	0.002	20
(2.6–3.2)				
21-crown-7 ^{c)} (308.4)	0.002	0.0398	0.002	20
(3.4–4.3)				
none	0	0.026	0.0013	20
propylene carbonate	0.0029	0.0145	0.0016	9.1

^{a)} All samples contained 10–25 mg of benzoyl peroxide as polymerization initiator.

^{b)} Estimated cavity diameters from ref. [6]. The radius of the Li^+ ion is 0.74 Å [13].

^{c)} The PVIC did polymerize in the normal way. However, the crown ether was not compatible and formed a separate layer.

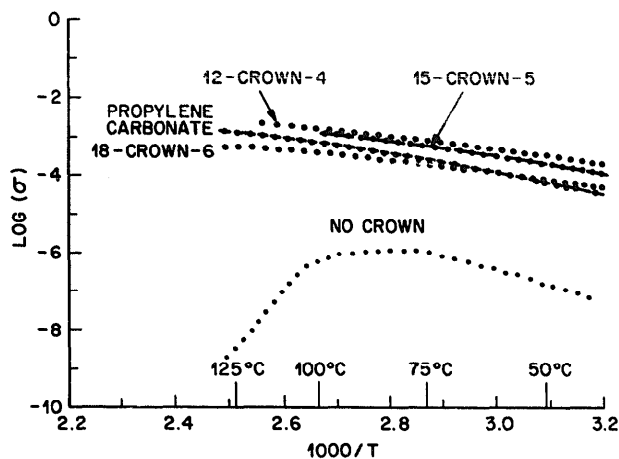


Fig. 1. Plot of $\text{Log } \sigma$ versus $1000/T$ for poly(vinylene carbonate) samples containing LiCF_3SO_3 and various crown ether ionophores.

ductivity by surrounding the cation with a large bulky group^{#1}. The data suggest, however, that the large enhancement factor that is observed may be analogous to the explanations proposed for ionic transport in biomembranes assisted by molecules such as valinomycin and gramicidin A [9]. These molecules, termed ionophores, are neutral molecules that provide an interior ligand cavity in which an inorganic cation may reside. The exterior of the ionophore is nonpolar thereby permitting the cation to move more easily in organic surroundings. The models proposed in biological systems include the action of such species as either carriers or as channels for transport. In the former case, the cation forms a complex with the carrier such as valinomycin, the complex moves across the membrane and then dissociates. In the latter case the transfer agent typified by gramicidin A remains in a fixed position in the membrane, with ions moving from site to site. We believe that a variation on the fixed site or channel model operates in our polymer films. In this case the fixed sites are the crown ethers which can move a little, but only in a narrowly confined region in the polymer matrix. The cations, assisted by motions of the polymer [10], could exchange themselves between nearest neighbor crown ether molecules. The directionality of the

^{#1} In our discussion of ionic transport we do not mean to imply that all the charge is carried by the cation. Charge transport by the anion may be substantial. See, for example, ref. [8].

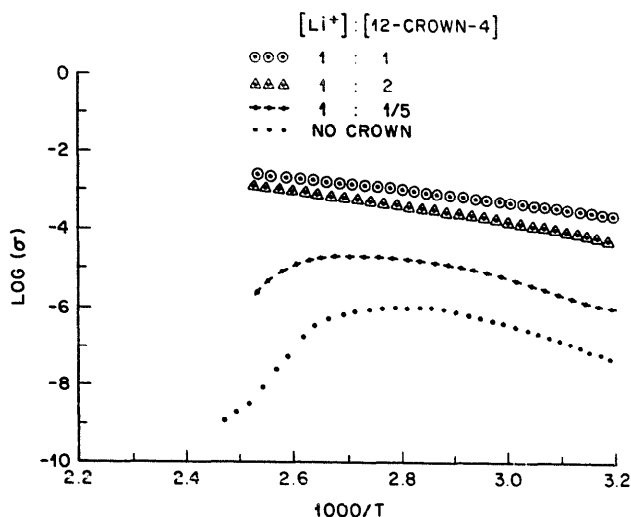


Fig. 2. plot of $\text{Log } \sigma$ versus $1000/T$ for poly(vinylene carbonate) containing the same concentration of LiCF_3SO_3 and varying amounts of 12-crown-4.

overall ionophoretic effect is determined by the field between the electrodes.

The effect of changing ionophore concentration is shown in fig. 2. Here we show conductivity plots for PVC and LiCF_3SO_3 mixtures containing varying amounts of 12-crown-4. When the ratio of lithium to crown ether is 1 the highest conductivities are obtained. When the crown ether is reduced to one-fifth the lithium concentration the conductivity is significantly lower. When twice as much crown ether is present the conductivity is not quite as high as when the ratio is one to one. In the high concentration sample the polymer was a little softer than the other samples. It is not known what roles, if any, this might play in our electrical measurement.

Finally, we note (fig. 1) that the conductivity versus temperature curves in the presence of crown ether follow smooth monotonic behaviour over the whole temperature range whereas, without the crown ether the conductivity shows a large bend at about 90°C . Whether the effect of the crown ether is only due to plasticization of the PVC and increasing salt solubility was of some concern to us. If propylene carbonate is added to the monomer together with Li^+ salt and the polymerization carried out in the usual way, a significant increase in conductivity is also obtained. This effect has been observed previously [11] in other polymers and believed to be due to plasticization. When small amounts of water are added

instead of propylene carbonate the conductivity is improved to even a greater extent ($\text{Log } \sigma = -2$ at 40°C) However, the $\text{Log } \sigma$ versus $1/T$ plots still have some curvature and further, the conductivity falls dramatically if the same sample is cooled and then remeasured. The use of crown compounds in PVIC significantly enhances ionic conductivity without producing thermal-history dependent data. Conductivities found with 12-crown-4 and 15-crown-5 are 5 to 10 times greater than with the addition of I, over the whole temperature range (fig. 1).

4. Conclusion

The addition of crown ethers to poly(vinylene carbonate) containing lithium salt causes a large increase in ionic conductivity. The largest effect is found with 12-crown-4, which is a specific lithium ion complexing agent. The conductivity at 40°C for the lithium trifluoromethanesulfonate complex is $2.5 \times 10^{-4} \text{ S cm}^{-1}$, the highest conductivity thus far reported in solid polymer electrolytes near room temperature. We believe that the data suggest a mechanism for charge transport of ionophore assisted ion motion [12] through the bulk polymer. We are currently investigating the electrical properties of other ionophore-containing polymer systems.

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